AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM275
Parallel and Distributed Systems

Final Examination
Fall Semester 2024/2025

Date: 9/Jan./2025

Number of Exam Pages: (3) (including this cover sheet)	Time Allowed: (2) Hours

Instructions:

- Total Marks: **50**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part I: Multiple Choice Questions (10 Marks, 2 marks for each)

1. Which of the following is a key benefit of using parallel computing?
   a) Reduced data access times
   b) Increased power consumption
   c) Higher cost of resources
   d) Sequential processing

2. In a distributed system, what is the role of middleware?
   a) To manage hardware resources directly
   b) To provide distribution transparency
   c) To perform network security tasks
   d) To create virtual machines

3. What is the main purpose of Docker in software deployment?
   a) To manage traditional virtual machines
   b) To provide an environment for developing web applications
   c) To create lightweight containers for applications
   d) To increase server security

4. In Flynn's Taxonomy, which category does SIMD belong to?
   a) Single Instruction, Single Data
   b) Multiple Instruction, Single Data
   c) Single Instruction, Multiple Data
   d) Multiple Instruction, Multiple Data

5. Which of the following is NOT a component of a distributed system?
   a) Client Nodes
   b) Middleware
   c) Docker Containers
   d) Server Nodes

Part II: Short Answer Questions (10 marks, 3+4+3)

1) Identify and explain the characteristics of a distributed system.

2) Compare the roles of a Docker image and a Docker container.

3) Discuss three advantages of using AWS for cloud computing.

Part III: Problem-Solving Question (30 marks)

Scenario: You are tasked with developing a student management system for a university that needs to track student information, course enrollments, and grades. The system will use a Cassandra database to manage data and Docker to deploy the application.

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1. **Design the Database Model** (3 marks):
   - Create a table named students to store student information, including fields for student_id, first_name, last_name, email, and enrollment_date.
2. **Create a Keyspace** (3 marks):
   - Define a keyspace named university with a replication factor of 3 and use Network Topology Strategy.
3. **Insert Sample Data** (3 marks):
   - Insert three records into the students table with sample student data (student id, first name, last name, email, enrollment date).
4. **Create a Course Table** (3 marks):
   - Create a table named courses to store course information, including course_id, course_name, and credits.
5. **Establish Enrollments Table** (3 marks):
   - Create a table named enrollments to link students to courses, including student_id, course_id, and enrollment_date.
6. **Insert Enrollment Records** (3 marks):
   - Insert records into the enrollments table for students enrolling in different courses.
7. **Dockerize the Application** (3 marks):
   - Write a basic Dockerfile to containerize the application for the student management system.
8. **Create Docker Compose File** (3 marks):
   - Write a docker-compose.yml file to define services for the student management application and a Cassandra database.
9. **Define a Query to Get All Students** (3 marks):
   - Write a CQL query to retrieve all students from the students table.
10. **Handle Docker Networking** (3 marks):
    - Explain how Docker networks will enable communication between your application and the Cassandra database.

End of questions

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